

Accessing Leiden via bus



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Make Leiden more accesible

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- ▶ Less mobile people
- ▶ Encourage bus use

Make Leiden more accesible

- ▶ Current system focusses on commuters
- ▶ Less mobile people
- ▶ Encourage bus use
- ▶ Small proportion

- Set of buslines L

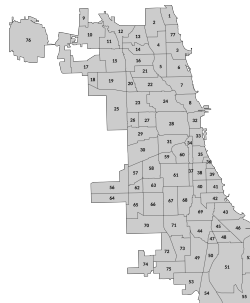


Figure: Chicago community areas

Rumpf and Kaul

- Set of buslines L
- Distribution of vehicles y

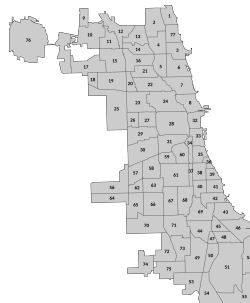


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Rumpf and Kaul

- ▶ Set of buslines L
- ▶ Distribution of vehicles y
- ▶ Origin & Destination points V^+, V^-

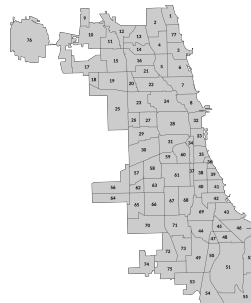


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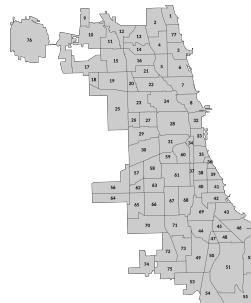


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Rumpf and Kaul

- Set of buslines L
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$$\max_y \text{Access}(y)$$

$$\text{s.t.} \quad \sum_{l \in L} y_l \leq \sum_{l \in L} y_l^*$$

$$y_l^{\min} \leq y_l \leq y_l^{\max} \quad \forall l \in L$$

$$y_l \in \mathbb{N} \quad \forall l \in L$$

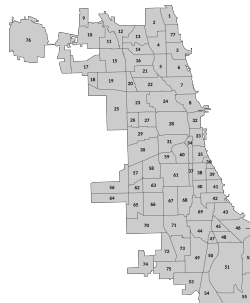


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$$y_l \in \mathbb{N} \quad \forall l \in L$$

$$x = \text{TransitAssignment}(y)$$

$$\text{OperatorCost}(y, x) \leq \text{OperatorCost}(y^*, x^*)$$

$$\text{UserCost}(x) \leq (1 + \varepsilon) \text{UserCost}(x^*)$$

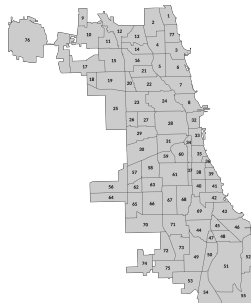


Figure: Chicago community areas

Access function

$$\text{Access}(y) = \sum_{\substack{\mathcal{K} \text{ minimum elements of} \\ \{A_i(y) | i \in \tilde{V}^+\}}} A_i(y)$$

$$A_i(y) = \sum_{j \in \tilde{V}^-} \frac{S_j d_{ij}^{-\beta}(y)}{F_j(y)} \quad i \in V^+$$

$S_j, F_j(y)$: destination attributes

$d_{ij}(y)$: travel time from i to j

$\beta > 0$, gravity decay parameter

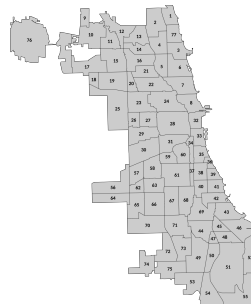


Figure: Chicago community areas

Choosing origins

$$\text{Access}(y) = \sum_{\mathcal{K} \text{ minimum elements of } \{A_i(y) | i \in \tilde{V}^+\}} A_i(y)$$

- Chicago regions too large

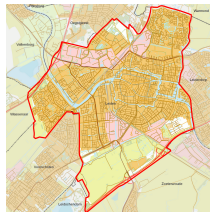
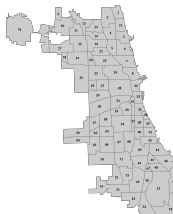


Figure: Leiden municipality

Choosing origins

$$\text{Access}(y) = \sum_{\mathcal{K} \text{ minimum elements of } \{A_i(y) | i \in \tilde{V}^+\}} A_i(y)$$

- Chicago regions too large

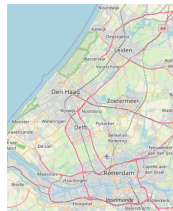
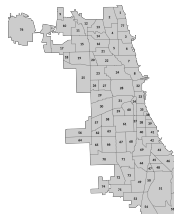


Figure: Part of the Randstad to scale

Choosing origins

$$\text{Access}(y) = \sum_{\mathcal{K} \text{ minimum elements of } \{A_i(y) | i \in \tilde{V}^+\}} A_i(y)$$

- Chicago regions too large
- Two methods

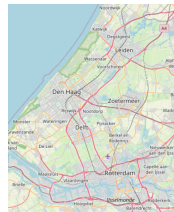
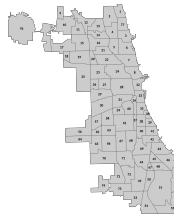


Figure: Part of the Randstad to scale

$$\text{Access}(y) = \sum_{\mathcal{K} \text{ minimum elements of } \{A_i(y) | i \in \tilde{V}^+\}} A_i(y)$$

► Postcode

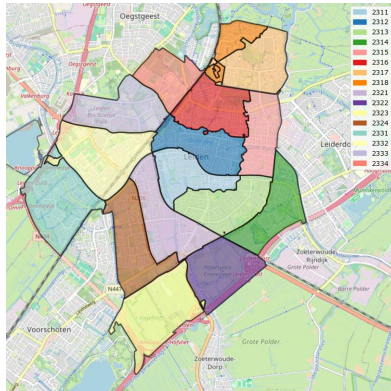


Figure: The different postcode regions of Leiden

$$\text{Access}(y) = \sum_{\mathcal{K} \text{ minimum elements of } \{A_i(y) | i \in \tilde{V}^+\}} A_i(y)$$

- Postcode
- Origin points

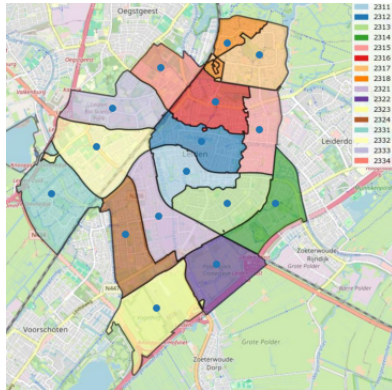


Figure: The different postcode regions of Leiden

$$\text{Access}(y) = \sum_{\mathcal{K} \text{ minimum elements of } \{A_i(y) | i \in \tilde{V}^+\}} A_i(y)$$

- ▶ Postcode
- ▶ Origin points
- ▶ Population known

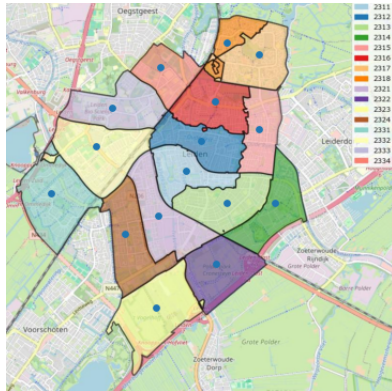


Figure: The different postcode regions of Leiden

$$\text{Access}(y) = \sum_{\mathcal{K} \text{ minimum elements of } \{A_i(y) | i \in \tilde{V}^+\}} A_i(y)$$

- ▶ Postcode
- ▶ Origin points
- ▶ Population known
- ▶ Still quite large

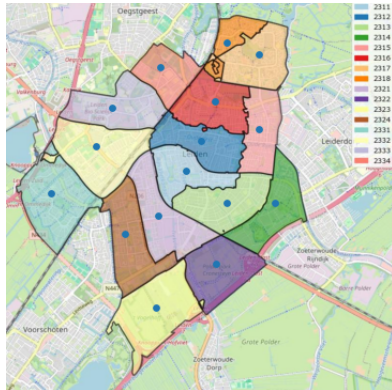
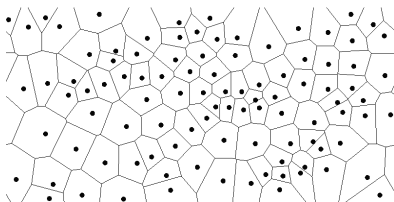


Figure: The different postcode regions of Leiden

Voronoi

$$\text{Access}(y) = \sum_{\substack{\mathcal{K} \text{ minimum elements of} \\ \{A_i(y) | i \in \tilde{V}^+\}}} A_i(y)$$

- ▶ Weight points
- ▶ Shortest distance



Voronoi

$$\text{Access}(y) = \sum_{\mathcal{K} \text{ minimum elements of } \{A_i(y) | i \in \tilde{V}^+\}} A_i(y)$$

► Busstops

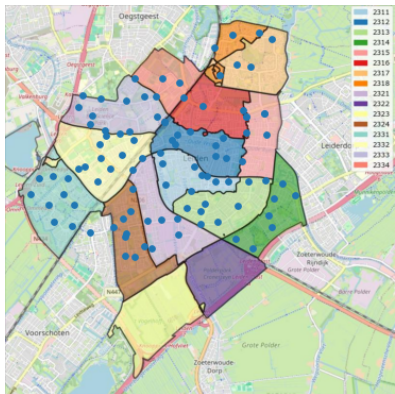


Figure: Busstops in Leiden

Voronoi

$$\text{Access}(y) = \sum_{\substack{\mathcal{K} \text{ minimum elements of} \\ \{A_i(y) | i \in \hat{V}^+\}}} A_i(y)$$

- ▶ Busstops
- ▶ Approximate populations
- ▶ Trips start at busstop

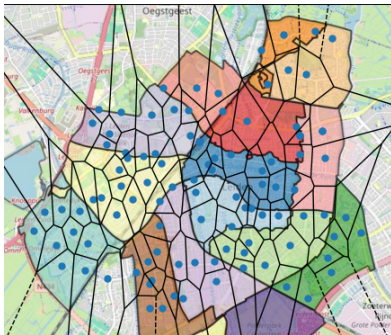


Figure: Busstops in Leiden

Destinations: GP locations

$$A_i(y) = \sum_{j \in \tilde{V}^-} \frac{S_j d_{ij}^{-\beta}(y)}{F_j(y)} \quad i \in V^+$$

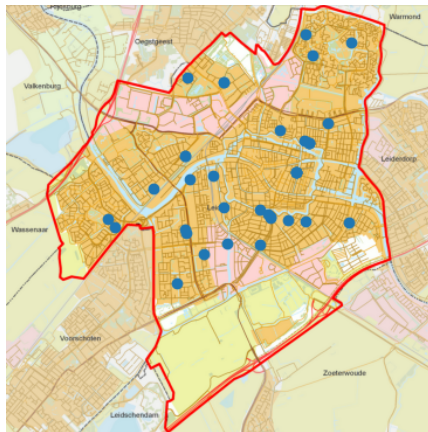


Figure: GP locations in Leiden

Destinations: GP locations

$$A_i(y) = \sum_{j \in \tilde{V}^-} \frac{S_j d_{ij}^{-\beta}(y)}{F_j(y)} \quad i \in V^+$$

S_j : facility quality

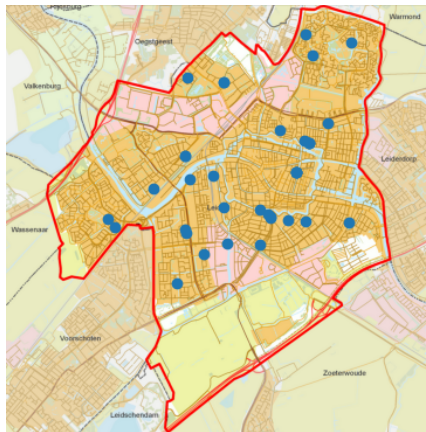


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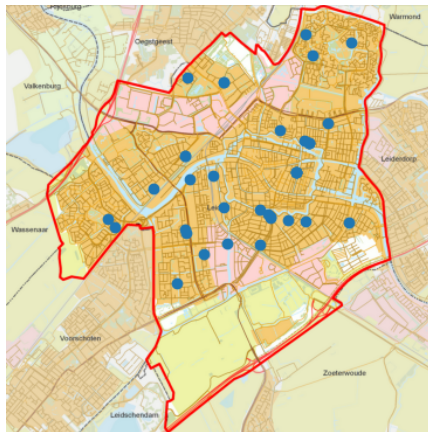


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P_i : population region i

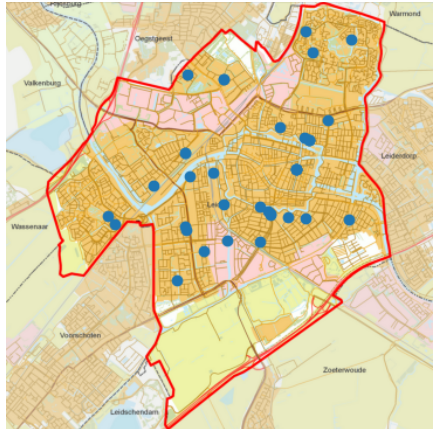


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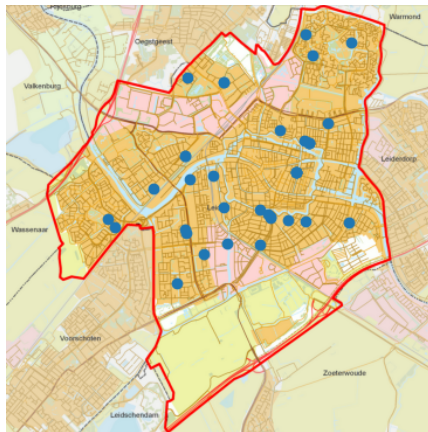


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$\beta > 0$, gravity decay parameter

► No hospitals

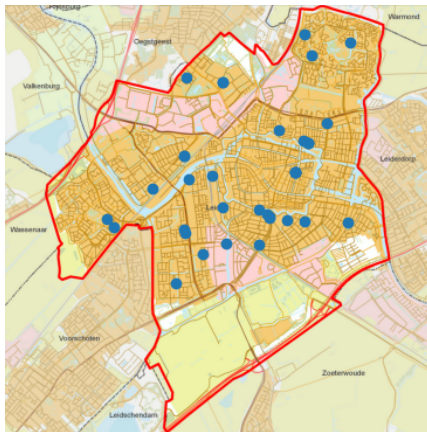


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Next steps

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 - ▶ Needed for $\text{UserCost}(x^*)$, $\text{OperatorCost}(y^*, x^*)$, $\text{TransitAssignment}(y)$

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- ▶ Run on PC4 regions

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